

Total No. of Questions : 5]

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F.Y. M.C.A. (Management)

MLT 562 MJ : MACHINE LEARNING TECHNIQUES

(2024 Pattern) (Semester - II)

Time : 2½ Hours]

[Max. Marks : 50

Instructions to the candidates:

- 1) *Scientific calculators are allowed.*
- 2) *All questions are compulsory.*
- 3) *Assume dataset wherever required.*

Q1) a) Differentiate between supervised and unsupervised learning? **[5]**

b) Explain logistic regression with suitable example. **[5]**

OR

c) Explain the concept of data set, labels and features. **[5]**

d) Explain dimensionality reduction techniques. **[5]**

Q2) a) i) A machine learning model is trained to predict if a person has cancer or not. It is tested on the dataset of 1000 patients. The model predicts 30 patients as having cancer and 970 to be healthy. Out of 30 cancer predictions, 20 are actually having cancer and out of 970 healthy predictions, 965 are actually healthy. Calculate accuracy and F1 - Score. **[5]**

ii) The following data shows the sales (in million dollars) of a company. Estimate the sales in the year 2020 using the regression line? **[5]**

x	2015	2016	2017	2018	2019
y	12	19	29	37	45

OR

P.T.O.

- b) A university wants to predict whether a student will pass or fail a course based on three factors : **[10]**

- i) Study Hours (High/low)
- ii) Attendance (Good/poor)
- iii) Previous Grades (Excellent/Average/Low)

The dataset is as follows:

Instance	Study Hours	Attendance	Previous Grades	Result
1	High	Good	Excellent	Pass
2	Low	Poor	Low	Fail
3	High	Poor	Average	Pass
4	Low	Good	Average	Pass
5	High	Good	Excellent	Pass
6	Low	Poor	Low	Fail
7	High	Poor	Low	Fail
8	Low	Good	Average	Pass

Construct a decision tree using the CART algorithm to predict the result of statement for the instance {Study Hours : High, Attendance: Poor, Previous Grades : Excellent}

- Q3) a)** Using the k-means algorithm and Euclidean distance, cluster the following 8 examples into 3 clusters. Consider A1, A4 and A7 as initial seeds (centroids).

A1 = (2, 10), A2 = (2, 5), A3 = (8, 4), A4 = (5, 8), A5 = (7, 5), A6 = (6, 4), A7 = (1, 2), A8 = (4, 9) **[10]**

OR

- b) Perform complete linkage agglomerative Hierarchical clustering using the given distance matrix.

Draw the dendrograms for clustering output which should clearly show the order in which the points are merged to form clusters: **[10]**

Item	A	B	C	D	E
A	0				
B	9	0			
C	3	7	0		
D	6	5	9	0	
E	11	10	2	8	0

Q4) Solve any two

- a) Given the following parameters for Q - learning problem: [5]
- Learning rate $\alpha = 0.2$
 - Reward $r_t = 10$
 - Current Q-value $Q(s_t, a_t) = 5$
 - Next state's Q-value $Q(s_{t+1}, a_{t+1}) = 8$
 - Discount factor $r = 0.9$
- How would you update the Q-value for state - action pair (s_t, a_t) ?
- b) Explain self - training and co-training techniques in semi - supervised learning. [5]
- c) Explain Deep Q-Networks (DQN) and compare DQN & Q - learning. [5]

Q5) Solve any one.

- a) A global bank faced significant financial losses due to credit card fraud. Despite traditional fraud detection methods, fraudulent transactions continued to slip through a machine learning - based fraud detection system to improve real - time fraud detection and reduce false positive. [10]
- Operations :
- i) Develop machine learning model in python for above problem.
 - ii) Identify independent variables and dependent variables.
 - iii) Explain the importance of data preprocessing before training the model.

OR

- b) A retail company operates multiple stores and wants to predicts daily/ weekly sales for each store based on historial sales data, store characteristics and external influences. [10]
- Consider the following dataset features : Sales, store ID, Day of the week, date perform following operations.
- i) Develop machine learning model in python for above problem.
 - ii) Identify independent variables and dependent variables.
 - iii) Explain the importance of data preprocessing before training the model.

